Lucio La Cava

PhD Student in Information and Communication Technologies

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Istruzione

PhD Student in Information and Communication Technologies

Rende, Italia

Dip. Ingegneria Informatica, Modellistica, Elettronica e Sistemistica (DIMES) - Università della Calabria *Novembre 2020 - Presente*

- **Advisor: Prof. Sergio Greco**
- **Co-Advisor: Prof. Andrea Tagarelli**
- **Topic: Data Science, Network Science, Machine Learning**

Laurea Magistrale in Ingegneria Informatica

Rende, Italia

Università della Calabria *Settembre 2017 - Luglio 2020*

- **Tesi: "Decentralizzazione di social network in federazione di istanze: modelli esplorativi e generativi di profilazione in Mastodon"**
- **Relatore: Prof. Andrea Tagarelli**
- **Valutazione finale: 110/110 con lode**

Laurea Triennale in Ingegneria Informatica

Rende, Italia

Università della Calabria *Settembre 2014 - Settembre 2017*

- **Tesi: "Outlier detection: tecniche per l'analisi dei dati"**
- **Relatore: Prof. Luigi Palopoli**
- **Valutazione finale: 106/110**

Esperienze

Esercitatore

DIMES, Università della Calabria

- **Data Mining**
  Corso di Laurea Magistrale in Ingegneria Informatica (Artificial Intelligence and Machine Learning) A.A. 2021/2022
- **Big Data Analytics**
  Corso di Laurea Magistrale in Computer Engineering for the Internet of Things A.A. 2021/2022
- **Networks and Systems in Bioinformatics**
  Corso di Laurea Magistrale in Ingegneria Informatica (Big Data and Bioinformatics) A.A. 2020/2021

Abilitazione all'esercizio della professione di Ingegnere

Rende, Italia

Sezione A - Ingegneria dell'informazione

Esame di Stato superato nella Seconda Sessione 2020

Reviewer

Conferenze (sub-reviewer)

- **KDD 2022 - 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining**
  August 14-18, 2022 - Washington DC (USA)
- **IJCAI 2022 - The 31st International Joint Conference on Artificial Intelligence**
  July 23-29, 2022 - Vienna (Austria)
- **NLDB 2022 - The 27th International Conference on Natural Language & Information Systems**
  June 15-17, 2022 - Valencia (Spain)
- **ACM The Web Conference 2022**
  April 25-29, 2022 - Lyon (France) - Virtual
- **Complex Networks 2021 - The 10th International Conference on Complex Networks and their Applications**
  November 30 - December 2, 2021 - Madrid (Spain)
- **ASONAM 2021 - The 2021 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining**
  November 8-11, 2021 - The Hague (Netherlands) - Virtual
- **SEBD 2021 - The 29th edition of the Italian Symposium on Advanced Database Systems**
  September 5-9, 2021 - Pizzo Calabro (Italy)

Journal

- **Social Network Analysis and Mining (SNAM)**
- **Information Systems (IS)**
- **Applied Network Science (APNS)**

Competenze

Macro-competenze	**Data Science, Network Science, Machine Learning, Artificial Intelligence, Cyber-Security**
Programmazione	Python, R, Java, Scala, Assembly, C, C++
Gestione Dati	Database relazionali, Database NoSQL, Database management systems (DBMS), Business Intelligence (OLAP)
Lingue	Italiano (Madrelingua), Inglese (Writing - C1, Reading - C1, Listening - B2, Speaking - B2)

Pubblicazioni

"LawNet-Viz: A Web-based System to Visually Explore Networks of Law Article References"
La Cava L., Simeri A., Tagarelli A.

Accepted for publication - The 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (ACM SIGIR 2022), Madrid, Spain, July 11-15 2022

Abstract: We present LawNet-Viz, a web-based tool for the modeling, analysis and visualization of law reference networks extracted from a statute law corpus. LawNet-Viz is designed to support legal research tasks and help legal professionals as well as laymen visually exploring the article connections built upon the explicit law references detected in the article contents. To demonstrate LawNet-Viz, we show its application to the Italian Civil Code (ICC), which exploits a recent BERT-based model fine-tuned on the ICC. LawNet-Viz is a system prototype that is planned for product development.

"Network analysis of the information consumption-production dichotomy in Mastodon user behaviors"
La Cava L., Greco S., Tagarelli A.

Accepted for publication - 16th International Conference on Web and Social Media (AAAI ICWSM-2022), Atlanta, Georgia, June 6–9 2022

Abstract: Decentralized Online Social Networks (DOSNs) are today an established alternative to the popular centralized counterparts. In this work, we push forward research on user behaviors in a decentralized context, by exploring the dichotomy between information consumption and production. Using the Mastodon user network as a proxy for the Fediverse landscape, we address two main research questions: Do the consumers, resp. producers, identified in one instance exhibit the same behavior consistently while interacting with other instances? and, Are there users who behave as consumers in one instance and simultaneously as producers in other instances, or vice versa? In this respect, our results reveal interesting traits of Mastodon users, yet unveil the emergence for further studies that can embrace other services in the Fediverse.

"The Italian Civil Code Network Analysis"
La Cava L., Simeri A., Tagarelli A.

Proceedings of the First International Workshop RELATED - Relations in the Legal Domain 2021 (RELATED 2021), São Paulo, Brazil, June 25, 2021

Abstract: The regulation of private law is a focal element in the metamorphosis of society, and by gaining a broader vision of the legal domain we might grasp novel or potentially hidden nuances and characteristics. In this work, we explore the Italian Civil Code (ICC) from an unprecedented perspective based on network analysis. We develop a text processing method to identify and extract article references from the ICC and, upon these, we define network models capturing their relation structures either at a book and corpus scale. The exploitation of the main structural features of these networks leads us to unveil meaningful patterns, holding within and across the books composing the ICC. Furthermore, by leveraging a community detection task, we investigate whether the formation of a community is related to the topic coherence of its assigned articles over the portions of books involved. Our findings reveal useful indicators that may help legal experts and practitioners enhance their knowledge from a novel perspective provided by the network of article references through the ICC.

“Understanding the growth of the Fediverse through the lens of Mastodon”
La Cava L., Greco S., Tagarelli A.

Applied Network Science 6, 64 (2021), Springer Open

Abstract: Open-source, Decentralized Online Social Networks (DOSNs) are emerging as alternatives to the popular yet centralized and profit-driven platforms like Facebook or Twitter. In DOSNs, users can set up their own server, or instance, while they can actually interact with users of other instances. Moreover, by adopting the same communication protocol, DOSNs become part of a massive social network, namely the Fediverse. Mastodon is the most relevant platform in the Fediverse to date, and also the one that has attracted attention from the research community. Existing studies are however limited to an analysis of a relatively outdated sample of Mastodon focusing on few aspects at a user level, while several open questions have not been answered yet, especially at the instance level. In this work, we aim at pushing forward our understanding of the Fediverse by leveraging the primary role of Mastodon therein. Our first contribution is the building of an up-to-date and highly representative dataset of Mastodon. Upon this new data, we have defined a network model over Mastodon instances and exploited it to investigate three major aspects: the structural features of the Mastodon network of instances from a macroscopic as well as a mesoscopic perspective, to unveil the distinguishing traits of the underlying federative mechanism; the backbone of the network, to discover the essential interrelations between the instances; and the growth of Mastodon, to understand how the shape of the instance network has evolved during the last few years, also when broading the scope to account for instances belonging to other platforms. Our extensive analysis of the above aspects has provided a number of findings that reveal distinguishing features of Mastodon and that can be used as a starting point for the discovery of all the DOSN Fediverse.

Talk/Conferenze

“Towards Mesoscopic Structural Analysis of the Fediverse of Decentralized Social Networks”
La Cava L., Ruffo L. E., Tagarelli A.

The 9th International Conference on Complex Networks and their Applications (CNA2020), December 1-3, 2020 – Madrid

Awards

2022	**Student Travel Grant**, ACM SIGIR 2022 - The 45th International ACM SIGIR Conference on Research and Development in Information Retrieval	*Madrid, Spain*